Do Districts That Spend More on Instruction Graduate More Students? California and Texas, 2018 Cohort

Ethan Tse · ECON 3916 Midterm Project · November 8, 2026

Repository: https://github.com/Ethan-Tse/econ3916-midterm

Notebook: https://colab.research.google.com/drive/1QSMaqcGo8LvYfKamGlujjVdNMRCJVtKn?usp=sharing

Question	Across regular school districts in California and Texas, how is a district's average per-pupil instructional spending over 2015–2018 (in 2018 dollars, logged) associated with its 2018 cohort graduation rate (the percent of ninth graders who graduate within four years), holding state and district size constant?
Headline finding	A district that spent 10% more per pupil over 2015–2018 (about $654 at the median) had a 2018 graduation rate 0.90 percentage points lower than a district in the same state and of the same size (95% CI −1.35 to −0.46 points, N = 1,366). This is a conditional association, not a causal effect: a missing confounder, district poverty and student need, can produce a negative sign by itself.
Data	Urban Institute Education Data Portal: EDFacts graduation rates (2018 cohort) and Common Core of Data finance and enrollment (2015–2018), downloaded 2026-11-05 and 2026-11-08; CPI-U from FRED. One row is one regular school district; N = 1,366 districts (963 Texas, 403 California).
Method	OLS of the graduation rate on log spending, a Texas indicator and log enrollment, with heteroskedasticity-robust (HC3) standard errors and t-based intervals.

1. The Question and the Data

State legislators and school boards decide every year how much to put into classroom instruction, and taxpayers pay for it. Whether districts that spend more per pupil graduate more of their students is the evidence behind those budget choices. I ask it as an association, because districts do not receive their spending at random. The outcome is the 2018 four-year graduation rate. The main explanatory variable is the average of a district's annual instructional spending per pupil over 2015–2018, the years that cohort was in high school, restated in 2018 dollars and logged so that a coefficient reads as the response to a percentage change in spending.

The data are a product of several measurement decisions. States' education agencies report the four-year adjusted cohort graduation rate to the U.S. Department of Education through its EDFacts system, a federal reporting requirement used for school accountability. It is the share of a district's first-time ninth graders (adjusted for transfers) who earn a regular diploma within four years. States set their own diploma requirements, so a 90% in California and a 90% in Texas need not mean the same thing, which is one reason state enters the model. To protect student privacy, small cohorts are reported as ranges, and the Education Data Portal supplies the midpoint of the range; that is why 105 districts, 99 of them in Texas, sit at exactly 75% (Figure 2). My download has graduation rates for 2018 only, so the analysis is one graduating cohort, not a panel.

Spending is current expenditure on instruction from the National Center for Education Statistics' Common Core of Data finance survey, which districts report through their states. It covers direct classroom costs and excludes support services, such as transportation and administration, and capital spending. I divide it by the district's total enrollment in the same year, so "per pupil" counts students of every grade while graduation depends only on high school. Each year is restated in 2018 dollars with the consumer price index for all urban consumers (CPI-U, FRED series CPIAUCNS, calendar-year average): prices rose 5.9% between 2015 and 2018, so a 2015 dollar counts as 1.059 dollars of 2018. A national index cannot adjust for price differences between California and Texas, which are left to the state indicator. I keep regular local school districts and drop County Offices of Education, special-purpose agencies with very small enrollments and, in 17 district-years across 7 offices, spending above $100,000 per pupil; 40 of them would otherwise be in the sample.

Of 1,958 districts in 2018, 1,371 report a graduation rate and 1,366 also have complete spending and enrollment; those 1,366 are the estimation sample. The 587 districts without a rate hide three different things. First, districts with no high school: 525 are in California, and 437 of those have "Elementary" in their name (a name-based check, not a certainty). They likely have no graduates, so the rate does not exist for them; they fall outside the population my question describes rather than being data to repair. Second, small districts: 36% of districts in the smallest enrollment fifth report a rate, against 73% to 87% in the other fifths (Figure 3), because privacy rules suppress or coarsen small cohorts. I label this MAR: whether a rate is reported depends mostly on cohort size, which I observe through enrollment and control for, so it is not MCAR. I cannot rule out an MNAR part. Rules of this kind can depend on the value of the rate itself, the data cannot test that, and the pile-up at 75% is its visible trace. Third, those 105 districts report only a range midpoint, not an observed rate. My estimates therefore describe districts with a high school and a reported rate, and the sample leans toward Texas (94% of its districts report, against 44% in California) and toward larger districts. Section 4 re-estimates without the 105 districts at 75%, and the coefficient stays negative. Table 1 shows each step from the full list to the sample, and Table 2 summarizes the 1,366 districts that remain.

Table 1. From the full list of districts to the sample

Step	Districts	What drops out
Regular local school districts, 2018 (County Offices of Education excluded)	1,958	
With a 2018 graduation rate	1,371	587 without a rate (no high school, or small-cohort suppression)
Also with complete 2015–2018 spending and enrollment (the sample)	1,366	missing spending or enrollment

Steps are applied in order, so a district missing two things is counted once, at the first step it fails.

Table 2. Summary statistics, 2018 cohort

Variable (units)	Mean	Median	SD	Min	Max	N
Graduation rate, 2018 cohort (%)	90.7	92.0	7.5	5.0	99.0	1,366
Spending per pupil, 4-year average, 2015–2018 (2018 $)	6,820	6,541	1,601	3,038	32,353	1,366
Enrollment, 2018 (students)	7,000	1,675	19,191	61	495,255	1,366

Variable (units)	Mean	Median	SD	Min	Max	N
Same spending average, not adjusted for prices ($)	6,611	6,344	1,555	2,990	31,654	1,366

Sources: Urban Institute Education Data Portal (EDFacts graduation rates, 2018 cohort; Common Core of Data finance and enrollment, 2015–2018), downloaded 2026-11-05 and 2026-11-08; BLS CPI-U via FRED. The last row repeats the spending row before the price adjustment. Enrollment is the 2018 count.

2. Descriptive Evidence

Most districts graduate nearly everyone (Figure 1): the median district has a 92% graduation rate and 74.7% are at or above 90%. The thin tail of low-graduation districts gives a skew of −3.18, because the rate cannot exceed 100. That skew comes from the ceiling, which a log would not fix, so I keep the rate in percentage points and check the residuals in Section 4. Spending is skewed the other way (skew 4.66 in dollars), so it enters the model as a log (skew 1.41), which lets a coefficient read as the response to a percentage change.

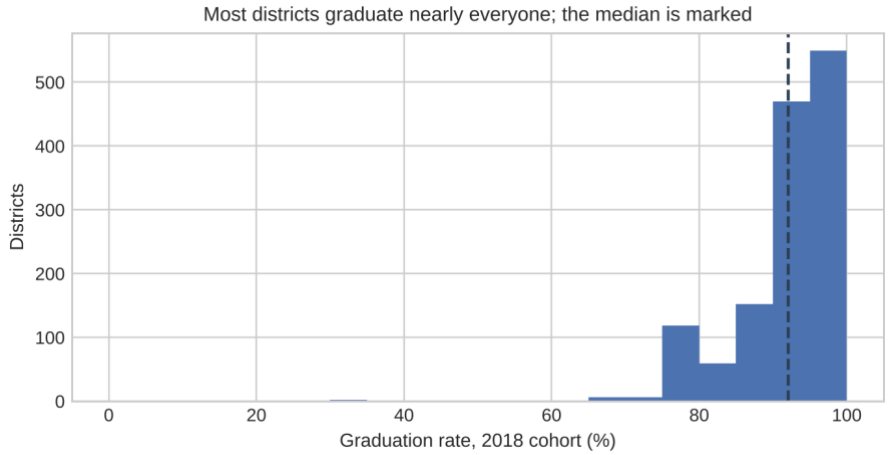


Figure 1. Most districts graduate nearly everyone: the median district (dashed line) has a 92% graduation rate, and 74.7% are at or above 90%. Source: Urban Institute Education Data Portal (EDFacts), 2018 cohort.

Before any controls, districts that spend more per pupil have slightly lower graduation rates, not higher (Figure 2): the correlation between log spending and the graduation rate is −0.32, about 1.21 points lower per 10% more spending. That is the opposite of what a spending-helps story predicts, which suggests the raw pattern reflects which kinds of districts spend more. The two states differ: California districts in the sample spend more ($7,723 against $6,442 on average, in 2018 dollars) yet graduate slightly fewer students (89.5% against 91.2%), and district sizes differ widely (median enrollment 4,770 against 1,098). State and size could therefore produce part of the negative association, which is why both enter the model. The flat band at 75% is 105 districts (99 in Texas) reported at a range midpoint, not a real cluster.

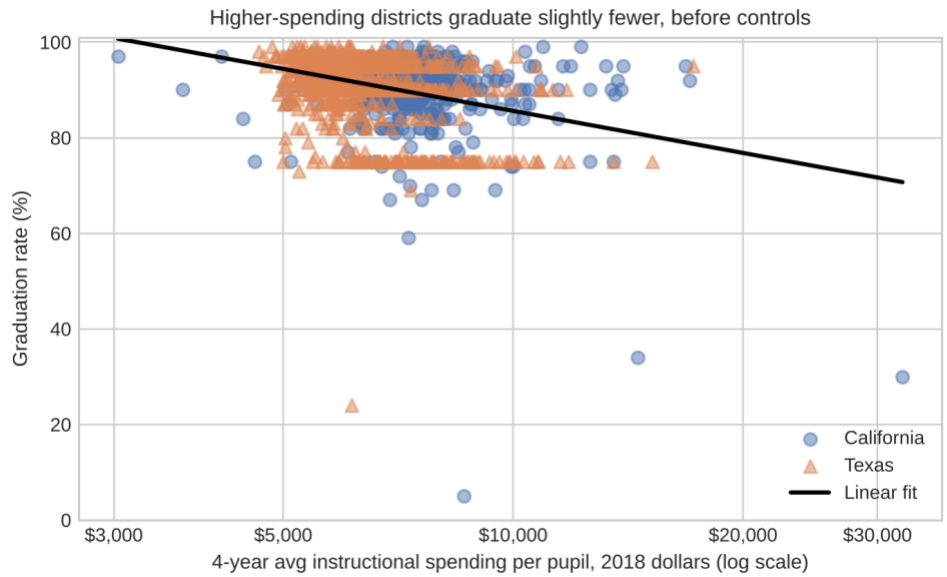


Figure 2. Before any controls, districts that spend more per pupil have slightly lower graduation rates: about 1.21 points lower for every 10% more spending (correlation −0.32). The vertical axis runs the full 0 to 100; the horizontal axis is logarithmic, so equal distances are equal percentage changes. Source: as Figure 1; Common Core of Data finance, CPI-U from FRED.

Figure 3 shows who has a graduation rate at all, which matters for what the results describe: Texas districts almost all report one, California districts mostly do not, and small districts report far less often than large ones.

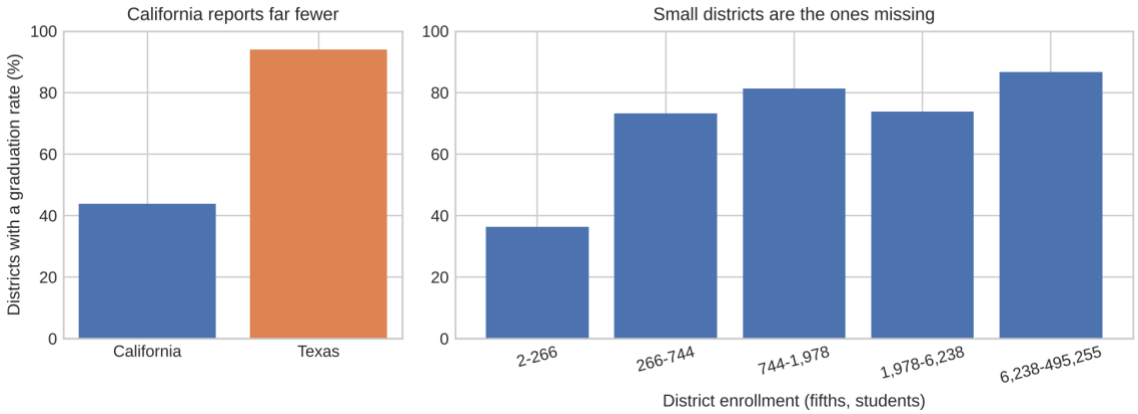


Figure 3. Graduation rates are reported for 94% of Texas districts but 44% of California's, and for 87% of the largest districts against 36% of the smallest. Source: as Figure 1.

3. The Model

I fit the following model by OLS, with heteroskedasticity-robust (HC3) standard errors and t-based intervals:

$$Graduation\ rate_i = \beta_0 + \beta_1\ ln(Spending_i) + \beta_2\ Texas_i + \beta_3\ ln(Enrollment_i) + u_i$$

Table A1 in the appendix gives the results; column 3 is the main model. **The spending coefficient** is −9.48 (robust standard error 2.36). Because spending is logged and the graduation rate is in percentage points, it reads in percent: a district that spends 10% more per pupil, about $654 more at the median of

$6,541, has a graduation rate 0.90 percentage points lower than a district in the same state and of the same size (95% confidence interval −1.35 to −0.46, $p < 0.001$). A one-standard-deviation difference in log spending, about 21% more spending, goes with a graduation rate 1.81 points lower, roughly a quarter (0.24) of one standard deviation of graduation rates. In plain terms: among otherwise similar districts, the ones that spend more graduate a slightly smaller share of their students. The difference is small, it is estimated tightly, and it runs opposite to what a spending-helps story predicts. That does not show that spending lowers graduation; Section 5 returns to why.

The enrollment coefficient is 0.94 (robust standard error 0.18). A district twice as large, with 100% more students, has a graduation rate 0.65 points higher than a district of the same state and spending level (95% confidence interval +0.41 to +0.89, $p < 0.001$). The difference is small, larger districts graduate somewhat more, and the interval stays well above zero.

Why each control is there. *State* (Texas = 1) blocks a confounding path. California and Texas differ in how schools are funded, in graduation requirements and in how rates are reported, all of which touch both spending and graduation. Adding it barely moves the spending coefficient (−12.69 to −13.17), so it earns its place on design grounds. It would be a bad control only if spending decided which state a district is in, which it cannot, so it is neither a mediator nor a collider. *Log enrollment* also blocks a confounding path: size affects spending per pupil (the correlation is −0.25, because small districts spread fixed costs over fewer students) and affects graduation (the correlation is 0.23). It does most of the work: adding it moves the spending coefficient from −13.17 to −9.48. That change of 3.69 is the omitted-variable bias formula at work: enrollment's own effect on graduation (0.94) times how much enrollment changes with spending (−3.90 per unit of log spending, because high-spending districts are small) gives −3.69, the gap between the short and the long regression. Enrollment would be a bad control if spending drew families into a district, because it would then sit on the path from spending to graduation (a mediator), or if it responded to both spending and unmeasured school quality (a collider). It is the 2018 count, measured after most of the spending, so I treat it as set mostly by the local population; columns 2 and 3 show the estimate without and with it, and both are clearly negative. *Left out:* district poverty and student need, the obvious confounder, since higher-need districts tend to receive more funding and to graduate fewer students. My data do not contain it. There is no year control because the sample is one graduating cohort.

4. Diagnostics

I ran the tests from Topic 14 on the main model; Figure A1 in the appendix shows the residual plots against the fitted values, against log spending and against log enrollment.

Residuals. The residuals sit around zero across the range of spending, with no bend, so a straight line in log spending is not obviously wrong. Their shape is lopsided: they fall as low as 81 points below zero but never rise more than 15 above it, because most districts are near 95% and a straight line can miss far downward but hardly upward, which is the 100% ceiling at work. The Jarque-Bera test rejects normal-looking residuals by a wide margin (JB = 27,547, $p < 0.001$; skew −2.86, kurtosis 24.2 against 3 for a normal distribution), which is that long lower tail. The band of points about 12 below zero is a measurement feature: all 105 districts reported at exactly 75% have residuals between −5 and −20, 98%

of them are among the smallest fifth of districts (median enrollment 167), and 75 is a range midpoint, not an observed rate.

Constant variance. The Breusch–Pagan test rejects constant variance (LM = 32.0, $p < 0.001$). The residual spread is 9.2 points in the smallest fifth of districts against 4.7 to 6.2 elsewhere, which makes sense: a small district graduates a small class, so its rate swings more. The 75% districts are not the whole story; leaving them out only brings the smallest-fifth spread down to 8.3. Unequal variance does not bias the coefficients, but it makes the usual standard errors wrong. For spending the classical standard error is 1.23 and the robust HC3 one is 2.36, 1.93 times as large; for the Texas indicator and enrollment the ratios are 1.20 and 1.23. That is why every model in Table A1 reports HC3 standard errors: the classical ones would have made the spending coefficient look nearly twice as precise as it is.

Collinearity. The variance inflation factors are 1.52 for log spending, 1.63 for the Texas indicator and 1.44 for log enrollment, far below the usual warning level of 5. The predictors overlap somewhat (size and spending have a correlation of −0.25, and the two states differ in both) but not enough to make the coefficients unstable.

Influence. No district comes close to the usual Cook's distance cutoff of 1 (the largest is 0.55), but 73 districts exceed the 4/N rule of thumb (0.0029), and 8 have a standardised residual beyond 3. Five districts (0.37% of the sample) carry 51% of the total Cook's distance. The top one shows why leverage matters. The Tuolumne County Superintendent of Schools (94 students, $32,353 per pupil, a 30% graduation rate) has a smaller residual (−6.2 standard deviations) than Laytonville Unified (−11.6), but a leverage of 0.055, 19 times the average of 0.0029, because its spending is far above every other district's. Dropping it alone moves the spending coefficient from −9.48 to −7.79, which is 18% of itself and 0.72 of its standard error, and takes the HC3 standard error from 2.36 to 1.71.

Two of the three most influential districts, Tuolumne and the Orange County Department of Education, are county-level agencies of the kind I excluded in Section 1; they differ only in name, so my name filter missed them. Table A2 shows the spending estimate under each exclusion. It stays between 0.62 and 0.97 points per 10% more spending, always negative and always with a 95% interval that excludes zero, so the sign does not depend on any one group of districts, but the size does, and I treat the main estimate as one point in that range. Deleting rows is right only for a reason that does not depend on the result, such as a record that is an error or lies outside the population being studied; a smaller standard error is never a reason. The two county agencies fit that reason on their face, but I found them through this check and not before, so I keep them in the main model and report the estimate without them beside it.

5. Threats and Conclusion

For the spending coefficient to be causal, districts that spend more would have to be otherwise the same as districts that spend less once state and size are held fixed. That is not true here, and the likeliest breaks run in opposite directions. First, need: districts with many low-income students often receive more money and graduate fewer students, and my data have no poverty measure, which can produce a negative coefficient with no effect of spending at all. Second, reverse causation: funding formulas can send more money to districts that are already struggling, so low graduation may lead to high spending

and not the other way round. The measures are also stand-ins: spending is per pupil across every grade, not what a high school student received, and a national price index cannot adjust for the different price levels in California and Texas; 105 small districts also enter at a range midpoint (Section 4). I checked what I could, and the sign survives dropping any group of influential or midpoint districts, but those checks cannot supply a confounder I did not measure. So my estimate is a **conditional association**: among California and Texas districts that have a high school and report a rate, those that spend 10% more per pupil graduate about 0.9 percentage points fewer students than districts of the same state and size. It is not the effect of raising a district's spending.

Conclusion. Higher-spending districts graduate slightly fewer students: 0.90 points per 10% more spending (95% interval −1.35 to −0.46), and across the checks in Section 4 the figure stays between 0.62 and 0.97. A reader can be fairly confident of the sign and rough size of this association, and should have no confidence that spending lowers graduation: the model explains 13% of the variation across districts, and the most important confounder is missing. With another month I would add district poverty and student need from the same portal, add more graduating cohorts so that each district can be compared with itself, and, if a funding reform changed some districts' budgets and not others, estimate its effect with a difference-in-differences. That design would buy a before-and-after comparison, and it would assume that the two groups of districts were on parallel paths until the reform.

References

Urban Institute. Education Data Portal: school districts, EDFacts graduation rates and Common Core of Data finance and enrollment. educationdata.urban.org. Downloaded 2026-11-05 and 2026-11-08.

Federal Reserve Bank of St. Louis. Consumer Price Index for All Urban Consumers: All Items in U.S. City Average, not seasonally adjusted (CPIAUCNS), from the U.S. Bureau of Labor Statistics. FRED. Annual averages of the monthly index, 2015–2018.

Appendix

Table A1. Graduation rate regressed on log spending and controls

	(1) Spending only	(2) Adds state	(3) Main: adds size
Log spending per pupil (4-year avg, 2018 $)	−12.69	−13.17	−9.48
	(1.95)	(2.14)	(2.36)
Texas (California = 0)		−0.49	1.28
		(0.50)	(0.63)
Log district enrollment			0.94
			(0.18)
Constant	202.45	207.04	166.19
	(17.12)	(18.92)	(21.70)
N	1,366	1,366	1,366
R^2	0.10	0.10	0.13

Note: Dependent variable: graduation rate, 2018 cohort (percentage points). Standard errors in parentheses: heteroskedasticity-robust (HC3), with t-based intervals. Column 3 is the main model. Column 1 has no controls; column 2 adds the state; column 3 adds district size.

Table A2. The spending coefficient under each exclusion

Version	Coefficient	Std. error	Points per 10% more spending	N
Main model (all districts)	**−9.48**	**2.36**	**−0.90**	**1,366**
Without the 2 county-level agencies	−6.74	1.35	−0.64	1,364
Without the 5 largest Cook's distances	−7.23	1.33	−0.69	1,361
Without the 105 districts reported at 75%	−6.53	2.98	−0.62	1,261
Without all 73 districts above 4/N	−10.18	1.17	−0.97	1,293

Note: Each row refits the main model on the stated districts, with HC3 standard errors. Every version is negative and its 95% interval excludes zero.

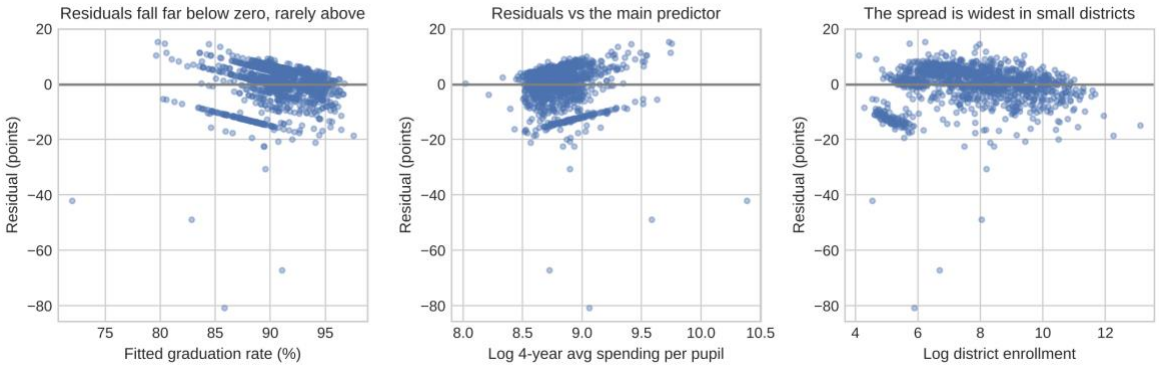


Figure A1. The main model's residuals are lopsided and widest among small districts: they fall as low as 81 points below zero and never rise more than 15 above it. The band about 12 below zero is the 105 districts reported at the 75% range midpoint. Residuals from the model with classical standard errors; the fit is the same as column 3.

Reproducing the numbers. Every number in this report is printed by the notebook's last code cell, "Numbers my report quotes," or by the cell that builds the table it appears in. The README gives the download step and the order to run the cells.